

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

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Topics covered :

1. Properties of Fourier Transform
2. Conditions for existence of Fourier Transform

- The **Fourier transform** of a signal $x(t)$ is defined as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (1)$$

and the **inverse Fourier transform** of the signal $X(j\omega)$ is defined as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (2)$$

- The Fourier Transform of the **impulse signal** $\delta(t)$ is given by

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = e^{-j\omega 0} = 1 \quad (3)$$

On the other hand, the Fourier transform of the **unit step signal** $u(t)$ is given by

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-j\omega t} dt \\ &= \frac{1}{-j\omega} e^{-j\omega t} \Big|_0^{\infty} = \frac{1}{-j\omega} \left[\underbrace{e^{-j\omega \infty}}_{\text{undefined}} - e^{-j\omega 0} \right] \end{aligned} \quad (4)$$

It is interesting to note that the FT of $u(t)$ is not defined ¹.

- The definition of the Fourier Transform clearly demonstrates that it is a linear operator. The properties of the Fourier Transform with respect to different operations performed on the input signal are as follows
 - **Time shift:** Let $X(\omega)$ represent the CTFT of the input signal $x(t)$. The FT of the delayed signal $y(t) = x(t - \tau)$ is given by

$$\begin{aligned} Y(\omega) &= \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t - \tau) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t_1) e^{-j\omega(t_1 + \tau)} dt_1 = e^{-j\omega \tau} X(\omega) \end{aligned} \quad (5)$$

A time shift in the input signal does not effect the magnitude of the Fourier Transform but causes the phase to change by $-\omega\tau$.

¹Link to an interesting article on Fourier Transform of a unit step signal: [link](#)

- **Time scaling:** Let $X(\omega)$ represent the CTFT of the input signal $x(t)$. The FT of a time scaled signal $y(t) = x(at)$ is given by

$$\begin{aligned} Y(\omega) &= \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt \\ &= \frac{1}{|a|} \int_{-\infty}^{\infty} x(t_1) e^{-j\frac{\omega}{a} t_1} dt_1 \quad \text{Note: } |a| \text{ ensures that limits are in order} \\ &= \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \end{aligned} \quad (6)$$

- **Differentiation:** Let $X(\omega)$ represent the CTFT of the input signal $x(t)$. The FT of a the signal $y(t) = \dot{x}(t)$ is given by [we derive this using the synthesis equation or the inverse Fourier Transform]

$$\begin{aligned} y(t) &= \frac{d}{dt} x(t) = \frac{d}{dt} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \right) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \frac{d}{dt} (e^{j\omega t}) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{j\omega X(\omega)}_{Y(\omega)} e^{j\omega t} d\omega \end{aligned} \quad (7)$$

$$\text{i.e. } Y(\omega) = j\omega X(\omega)$$

- **Duality Property:** Let $X(\omega)$ represent the CTFT of the input signal $x(t)$. The FT of a the signal $X(t)$ (ω replaced by t) is given by [start from synthesis equation]

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad \text{Now, let } t_1 = -t \\ 2\pi x(-t_1) &= \int_{-\infty}^{\infty} X(\omega) e^{j\omega t_1} d\omega \quad \text{Change the variable } \omega \text{ to } t_1 \\ 2\pi x(-\omega) &= \int_{-\infty}^{\infty} X(t_1) e^{j t_1 \omega} d\omega \\ 2\pi x(-\omega) &= F\{X(t)\} \end{aligned} \quad (8)$$

- The area under the curve of the non-periodic signal can be computed from the Fourier Coefficient at $\omega = 0$ as

$$X(\omega)|_{\omega=0} = \int_{-\infty}^{\infty} x(t) e^{j0t} dt = \int_{-\infty}^{\infty} x(t) dt = \text{Area} \quad (9)$$

On the contrary, the area under the curve of the Fourier Transform of $x(t)$ i.e. $X(\omega)$ can be computed from the value of $x(t)$ at $t = 0$. To illustrate that consider the synthesis equation at $t = 0$ i.e.

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega 0} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) d\omega = \frac{1}{2\pi} (\text{Area}) \quad (10)$$

- **Convergence of Fourier Transform:** Consider a signal $\hat{x}(t)$ that is synthesized using the Fourier Transform

$X(j\omega)$ (obtained using the analysis equation on original signal $x(t)$) i.e.

$$\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (11)$$

The conditions under which the synthesized signal $\hat{x}(t)$ approaches the original signal $x(t)$ are called the convergence conditions. There are two set of convergence conditions reported in the literature.

- Convergence in Mean Square Error/Power sense: If the signal $x(t)$ is square integrable i.e.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty \quad (12)$$

then it can be proven that the mean square error (MSE) between the signals $x(t)$ and $\hat{x}(t)$ is zero i.e.

$$\int_{-\infty}^{\infty} |x(t) - \hat{x}(t)|^2 dt = 0 \quad (13)$$

It must however be noted that a zero MSE does not mean that the signal $x(t)$ and $\hat{x}(t)$ are exactly identical. The MSE convergence indicates that the two signals carry the same power.